
MSc Viva Cheat-Sheet

Opening Elevator Pitch (90s)

- **Problem:** Healthcare queries mix **structured biomarkers** and **unstructured text**. A single model is inefficient. We need a **lightweight dynamic router** that directs each query to the right module under strict **p95 ≤ 100 ms latency, ≤ 2 GB RAM**.
 - **Constraints:** Text-only routing, calibrated probabilities, abstention/hedging, reproducibility.
 - **Contributions:**
 1. **Calibrated, text-only router** (temperature scaling, optional LLM prior, abstention/hedging).
 2. **Four-module orchestration** (Prediction, CBR, QA, Summarisation) with consistent interfaces and explanations.
 3. **Full reproducibility harness:** fixed splits, seeds, artifacts, logs.
 - **Headline results:**
 - Router **Accuracy 0.9668** (95% CI [0.9566–0.9747]); ECE $\downarrow$ post-calibration; p50 18–24 ms; p95 70–95 ms.
 - Prediction: Synthetic Macro-F1 ~ 0.99 , Hybrid ~ 0.70 .
 - CBR: Top-1 0.78–0.85, Top-3 up to 0.92, 20–70 ms.
 - QA: EM 0.45–0.55, F1 0.55–0.65; unsupported claims $\downarrow \sim 40\%$ via retrieval gate.
 - End-to-end: DLQR outperforms “always-one-module” baselines at matched budgets.
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General / Background

Q1. Gap & RQs?

A: Existing cascades/static routers can’t meet strict latency/memory budgets. My work introduces a calibrated, text-only router with abstention/hedging to dynamically select modules. Four RQs:

1. Can DLQR improve task utility vs static baselines?
2. How accurate, calibrated, and fast can routing be?
3. How do individual modules perform under the framework?
4. How reproducible are results and what are the threats?

Q2. Three main contributions?

A: (1) Calibrated text-only router with abstention/hedging and optional LLM prior.
(2) Orchestration of four modules with interpretability.
(3) Full reproducibility harness with artifacts and decision logs.

Methods / Design Choices

Q3. Why temperature scaling?

A: It adjusts logits with τ to calibrate probabilities. Accuracy stays unchanged, but ECE and NLL drop, making thresholds reliable. Validation gave $\tau \approx 0.992$, $\tau_{\text{mix}} \approx 1.0$.

Q4. When abstain vs hedge?

A: If calibrated confidence $< \text{TU}$ (0.5–0.8), router abstains. If two candidates are close under high uncertainty, router hedges by passing both. LinTS is optional for safe online exploration.

Q5. Router metrics & latency tails?

A: Accuracy 0.9668, ECE reduced, p50 18–24 ms, p95 70–95 ms. Tails come from retrieval/generation in QA/Sum, not the router itself.

Q6. Why synthetic $\gg$ hybrid performance?

A: Synthetic data are well-separated by design $\rightarrow \text{F1} \sim 0.99$. Hybrid is sparse and imbalanced $\rightarrow \text{Macro-F1} \sim 0.70$, AUROC ~ 0.77 . Calibration improves thresholds but highlights real-world difficulty.

Q7. How explainability is handled?

A: Prediction uses SHAP. CBR shows nearest neighbours and prototype rationales. QA returns evidence citations. Summarisation constrained to retrieved text.

Q8. CBR 100% vs 0.78–0.85 — why?

A: Ablation “Text-Only” ran in a controlled setup, yielding inflated scores. Full measured evaluation added noisy BCM re-ranking, reducing accuracy. Unified fusion balanced the trade-off.

Q9. CBR fusion & signals?

A: Fusion is α -weighted blend of text similarity and BCM scores. Module outputs retrieval quality and coverage to inform router decisions.

Q10. QA safety?

A: Hybrid retrieval + cross-encoder. A retrieval-confidence gate filtered weak evidence, reducing unsupported claims by $\sim 40\%$ with small recall cost.

Data / Experiments

Q11. End-to-end comparison (RQ1)?

A: DLQR achieved higher Macro-F1/AUROC than “always-Prediction” or “always-QA,” while staying within $\text{p95} \leq 100$ ms routing budget. Latency tails came from retrieval/generation.

Q12. Metrics & CIs?

A: Accuracy, Macro-F1, AUROC/AUPRC; calibration metrics (ECE, Brier, NLL); latency p50/p95; throughput; cost proxy. Wilson intervals for proportions; stratified bootstrap for F1/AUROC/AUPRC (1,000 resamples).

Q13. How reproducibility ensured?

A: Fixed splits, preprocessing only on train, fixed seeds, profiling harness, bootstrap evaluator, archived artifacts with checksums, and JSONL decision logs.

Results / Analysis

Q14. Why text-only router?

A: Enforces simplicity and speed, avoids leakage of structured features, and respects $p95 \leq 100$ ms / ≤ 2 GB constraint. Modules handle complexity downstream.

Q15. Effect of α and w ?

A: $\alpha=0.7$ balanced text and numeric signals in CBR; BCM-only hurt accuracy. w mixing LLM intent priors (0.3–0.7) showed stable performance; post-hoc temperature scaling ensured calibration.

Future Work & Impact

Q16. Two threats to validity?

A: Synthetic/hybrid data may not reflect real patients → mitigated with hybrid dataset. Calibration may drift under distribution shift → mitigated with bootstrap CIs, logs, and future conformal layers.

Q17. Safe online updates?

A: Log all routing decisions; conservative LinTS with priors; rollbackable configs; maintain strict abstention until enough evidence supports updates.

Q18. Cost proxy & role?

A: Proxy combined CPU ms and token usage. Router used it for tie-breaking, avoiding expensive modules when marginal utility was low.

Q19. Why separate QA & Summarisation?

A: QA enforces evidence grounding; Summarisation handles synthesis and rewrites safely. Both return adequacy/confidence scores to router.

Q20. Most impactful next step?

A: Conformal acceptance for distribution-free calibration, plus evaluation on more realistic clinical datasets while staying within system budgets.

Likely Criticisms & Defenses

- **Router too limited (text-only):** Feature, not flaw — ensures speed, avoids leakage, and respects strict compute budgets.
- **Hybrid prediction weak (~0.70 F1):** Reflects real-world difficulty; router can defer or abstain; calibration helps safe use.
- **CBR inconsistency (100% vs ~0.8):** Ablation vs measured differences show why naive BCM hurts; fusion is more realistic.
- **Latency tails (QA/Sum):** Router $p95 \leq 100$ ms; tails come from retrieval/generation, which is expected and quantified.
- **Synthetic data validity:** Acknowledged; mitigated with hybrid datasets; future real-world evaluation planned.
- **Distribution shift risk:** Future work includes conformal calibration and safe online learning with audit trails.
- **Ethical/clinical scope:** Clearly stated as software/system contribution; not making clinical claims.